PIP-II Beam Transfer Line - Dipole Magnets Statement of Work

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Document Approval

Signatures Required	Date Approved
Originator: Dino Karas	As in Teamcenter
Reviewer: Jeremiah Holzbauer	As in Teamcenter
Reviewer: Jeremy Nudell	As in Teamcenter
Reviewer: Tom DiGrazia	As in Teamcenter
Approver: Daniel Jones	As in Teamcenter

Revision History

Revision	Date of Release	Description of Change
-	February 2026	Initial Release
A	March 2026	Removed Section 9 regarding warranty. Updated Technical Requirements section (previously Section 11 titled Bid Evaluation Criteria). Updated drawing revisions in Section 12.

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1. INTRODUCTION

The Proton Improvement Plan Phase 2 (PIP-II) is an essential upgrade to the Fermi National Accelerator Laboratory (FNAL) accelerator complex, which will provide high-intensity proton beams for FNAL's broad physics program. Figure 1 illustrates the overall layout of the accelerator complex that will be upgraded or newly constructed as part of the PIP-II project. The PIP-II Linear Accelerator (Linac) is a new Superconducting Linac designed to accelerate the H- charged particle beam to 800 MeV. The Beam Transfer Line (BTL) is designed to transport beam from the Linac to the Booster injection region where it will further increase in energy. The BTL is comprised of 2 main transfer arcs and a straight section which can either direct the beam into the absorber in the beam abort line or into the Booster ring.

Dipoles are used to transport the H- charged particle beam within the BTL from the end of the Linac to the Booster. There will be a total of 39 dipoles with 3 distinct varieties installed in the BTL. The specifics of these are outlined within this Statement of Work (SOW).

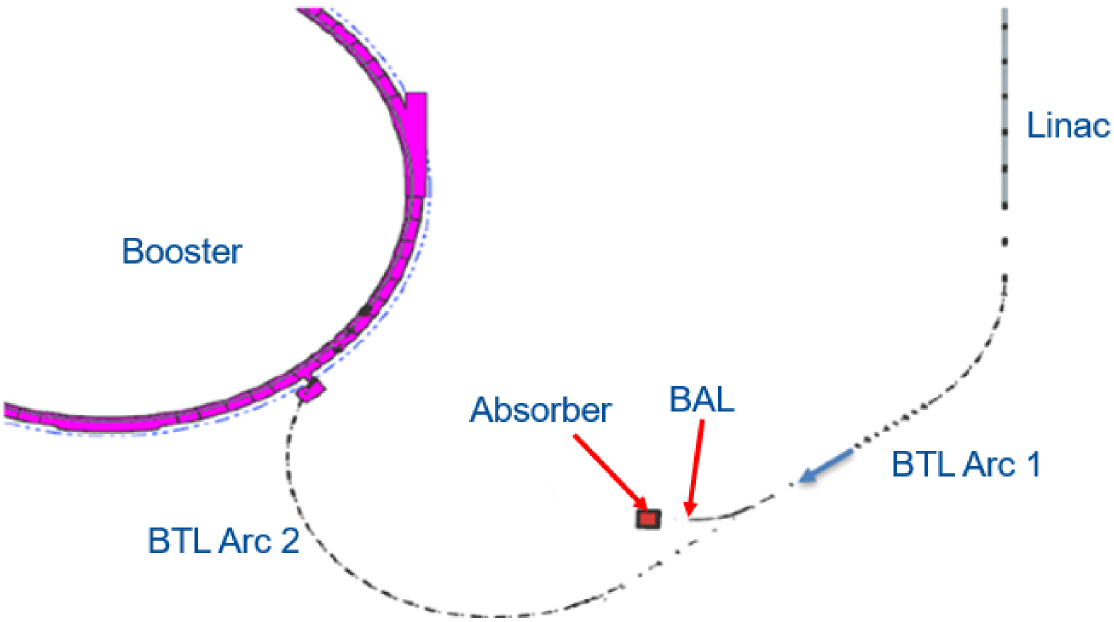

Figure 1: General Layout of PIP-II Accelerator Complex Upgrades

This SOW outlines the technical tasks and requirements for the Prospective Offerors (hereafter referred to as "the Subcontractor"), including reporting, documentation, scheduling, change management, deliverables, and communication with The Buyer, Fermi Forward Discovery Group (FFDG). These requirements pertain to the various type of BTL dipoles and must be fulfilled in accordance with FNAL standards and guidelines.

2. SCOPE AND DELIVERABLES

The primary scope of work covers design, manufacturing, fabrication, quality control (QC), magnetic testing, and delivery for a build-to-spec package inclusive of all associated and/or described work within this SOW as it relates to the BTL dipoles. The Subcontractor will provide all shop facilities, manufacturing/fabrication equipment, qualified shop personnel, engineering personnel, inspection services, magnetic testing services, cleaning services, and packaging services for completing the scope of work defined in this SOW.

Table 1 and Table 2 provides a list of all contract deliverables for the first articles and optional production units respectively, including documentation which is listed as required (AR). FFDG reserves the right to exercise an option for the production units which is contingent on the first articles meeting the requirements in Section 3 as verified through the Site Acceptance Tests (SATs) outlined in Section 8.2. Only released drawings shall be used for manufacturing and fabrication. The Subcontractor is responsible for verifying the drawing release state with FFDG prior to the start of any manufacturing or fabrication. The requirements for documentation to be provided to FFDG are outlined in Section 7.

Table 1: List of First Article Contract Deliverables

Item	Quantity
Regular Dipole First Article	1
End of Line (EOL) First Article	1
Regular Wide Dipole First Article	1
Regular Dipole First Article Complete Drawing Set	AR
EOL Dipole First Article Complete Drawing Set	AR
Regular Wide Dipole First Article Complete Drawing Set	AR
First Article Production Schedule(s)	AR
Material Certifications of Major Components	AR
Final Design Report	AR
QC: Fabrication & Inspection Plans	AR
Magnetic Measurement Test Plans	AR
QC Reports: Fabrication & Inspection Report	AR
Magnetic Measurement Test Reports	AR

Table 2: Optional Production Contract Deliverables

Item	Quantity
Regular Dipole Production Units	39
EOL Dipole Production Units	3
Regular Wide Dipole Production Units	2
Complete Drawing Set for All Magnet Types	AR
Production Units Production Schedule	AR
Material Certifications of Major Components	AR
QC: Fabrication & Inspection Reports	AR
Magnetic Measurement Test Reports	AR

3. REQUIREMENTS/SPECIFICATIONS

The design of the regular and regular wide dipole magnets shall be a straight, H-Frame magnet design. The design for the EOL dipoles shall be straight, H-Frame or Window Frame magnet design. The magnets must be splitable to facilitate the installation of the beam tube. The drawings and .step files of the beam tubes can be found in the attachments. The design must ensure that the magnet will maintain all magnetic properties, including magnetic axis positions, within the stated specifications upon disassembly and subsequent re-assembly. Table 3 and the remaining information in this section outline the overall design requirements which must be considered when designing the dipole magnets of each type. The specific design requirements found in Section 3.1, 3.2, and 3.3 for each magnet type supersede the overall design requirements should contradictory statements be found during the preliminary design stage.

Table 3: Overall Dipole Design Requirements

Requirement #	Requirement Statement
T-D-121.03.05-D001	All external water connections shall be ½" Male National Pipe Thread (MNPT) Swagelok fittings
T-D-121.03.05-D002	Maximum water pressure drop across the magnet shall be less than 4.14 Bar (60 psid)
T-D-121.03.05-D003	Minimum water pressure drop across the magnet shall be greater than 0.344 Bar (5 psid)
T-D-121.03.05-D004	Water flow velocity in the magnet coils shall be between 0.91 to 3.05 m/s (3 to 10 ft/s)
T-D-121.03.05-D005	Outlet water temperature rise shall not exceed 10°C (18°F) above the nominal inlet temperature.
T-D-121.03.05-D006	The water flow rate shall be less than 11.73 L/min (3.1 G/min)
T-D-121.03.05-D007	The maximum allowable working pressure shall be 10.3 Bar (150 psi)
T-D-121.03.05-D008	The coils shall have a minimum water passage diameter of 5 mm (0.2")
T-D-121.03.05-D009	Maximum magnet Temperature must not exceed 46.1°C (115°F)
T-D-121.03.05-F001	All magnets shall allow for mounting to a 3-point alignment system
T-D-121.03.05-F002	All magnets shall have provisions to allow for rigging the magnet using swivel hoist rings meeting ASME B30.26 requirements
T-D-121.03.05-F003	The maximum weight of each magnet shall not exceed 2268 kg (5000 lbs.)

The magnet cores shall be constructed from non-grain oriented, electrical steel laminations. The electrically isotropic laminations shall have a C3 insulative coating conforming to the requirements in ASTM A976-03. A bondable coating on the laminations to bond the stacked laminations together is permitted. To minimize the hysteresis, the laminations shall have a maximum thickness of 1 mm (0.039") and have a coercivity, Hc, less than 60 A/m. The laminations must meet all magnetic properties of each magnet type outlined in this SOW. All laminated steel master coils used to fabricate the production unit laminations shall be from the same batch of steel supplied from a singular vendor. Fermilab shall review and approve the laminated steel material and method used to fabricate the laminations. The core endplates must be rigid to not allow for bowing of the core. Threaded rods or similar secured to the endplates shall pass through the core to increase the structural stability.

Each magnet must incorporate grounding plate. The plate shall be a minimum of 76.2 mm x 76.2 mm x 25.4 mm thick (3" x 3" x 1" thick) with two ½"-13 threaded holes separated by 44.45 mm (1.75") to accommodate one MCM 500 two-hole Burndy YA342N lug. It shall be fixed to top of the magnet near the center with respect to the magnet length. The top surface of the grounding plate mating with the Brundy lug shall be tin plated. The bottom mating surface of the grounding plate shall be in contact with an unpainted surface on the magnet body.

Each magnet shall have provisions to allow for rigging the magnet using swivel hoist rings meeting ASME B30.26 requirements. The swivel hoist rings shall be located about the center of gravity of each magnet type and properly sized/rated for the load. Each swivel hoist ring must be marked to display the name or trademark of manufacturer, related load, and torque value. The Subcontractor is obligated to provide swivel hoist rings for a minimum of 10% of the total magnets including all first article magnets. The placement of the eyebolts must be reviewed and approved by FFDG.

The conductors shall be made from Oxygen Free, High Conductivity (OFHC) C10200 copper alloy with electrical conductivity of 100% IACS per ASTM B188. The conductors shall be bright annealed and air cooled to a dead soft temper (maximum Rockwell F-45). The OFHC copper shall be free of surface defects, damage, nicks, cuts, and/or scratches. The uniformity of the conductor shall be such that the resistance of all coils constructed from it shall be equal to ±1%. The conductor shall have a minimum water passage diameter of 5 mm (0.2”). The wound coils shall be such continuous runs such that there are no joints permitted within the water passage of the coils. Finished coils shall be insulated with fiberglass tape and vacuum impregnated with epoxy. The leads shall also be insulated with fiberglass and coated in epoxy. Unsupported epoxy volumes greater than 1.59 mm (1/16”) in thickness shall be deemed unacceptable.

All current carrying components (jumpers, flags, etc.) for each magnet shall be manufactured from OFHC copper. Flags for connecting the magnets to the power supplies shall have thru holes separated by 44.45 mm (1.75”) for ½”-13 bolts and designed to accommodate 2 Burndy YA342N lugs for MCM 500 cables on each flag. The flags shall be electrically isolated from the magnet bodies with G10. All electrical connections shall either be brazed or fixed with bolts and Bellville washers. Lock washers are permitted where Bellville washers cannot be used. The coil connection posts, jumpers, exposed bus work, and all metallic parts connected to them shall be protected against accidental contact by an insulating, transparent cover which can only be removed with tools.

The inlet water temperature is 35°C ± 2.8°C (95°F ± 5°F) under normal operating conditions. Since the magnets will be installed in an underground tunnel with no temperature control, experience shows the air temperature can vary between 23.9°C to 37.8°C (75°F and 100°F). Therefore, heat dissipated to the air should be minimized. The water circuits must be connected in parallel and connect to a common inlet and outlet manifold. Each magnet shall have one supply and one return manifold which connects to the main low conductivity water (LCW) system. The inlet and outlet manifolds shall face upwards (i.e., no downward facing taps). The manifolds must be electrically insulated from the coils using PEEK, G10, or ceramic connectors between the coil and the manifolds. The external water connections for the manifolds shall be ½” Male National Pipe Thread (MNPT) Swagelok fittings. All internal water connections shall be made with appropriate Swagelok fittings. All fittings shall conform to the standards defined by ANSI/ASME B1.20.1. All manifolding and water circuits must be constrained to not allow for motion that could result in damage. Thermal switches shall be installed on each water circuit using thermally conductive epoxy. The thermal switch specification sheet and installation location must be reviewed and approved by FFDG.

The magnet must include the ability to mount the magnets onto a 3-point alignment system. One or more support bars shall be mounted to the bottom surface of the lower half core. The bar(s) shall incorporate either thru holes for ½-13 bolts or ½-13 threaded holes to accept bolts for mounting the magnet to the alignment system. The bar(s) thickness and mounting location must be sufficient to support the full magnet weight. The hole type and locations on the support bars shall be provided by FFDG upon release of the alignment system design after the final magnet dimensions and weight have been established.

Each magnet must accommodate for a minimum of 8 1.5TH Series Spherically Mounted Retroreflector (SMR) adapters. The adapters which accept 1.5” diameter retroreflectors require a 6.35 mm (0.25”) diameter hole with a minimum depth of 12 mm (0.47”) and a 42 mm (1.65”) diameter spot face. At a minimum, each core endplate shall be designed to accommodate the adapters at the corners on top and bottom of the end plates (4 on each end of the core).

The finished magnets shall be painted NAL Light Blue (Specs RGB:65,182,230, HEX: #41B6E6, CMYK: 67,2,0,0, PMS: 298C 298U). A nameplate shall be permanently fixed onto the magnet indicating the design parameters and/or measured data such as resistance, inductance, etc. as well as the magnets physical characteristics. Each nameplate shall also contain a unique serial number as stated in Section 8.3 to identify the magnet. The information required to be displayed on the nameplate shall be provided by FFDG.

3.1. Regular Bend Dipoles

The regular bend dipoles will be powered in either strings of 5 magnets or a string of 20 magnets. All magnets should be interchangeable in the circuit configuration and, as such, are required to have an identical integrated BdL fields within ±0.1% (this includes when a set of magnets is turned off and then set back to their nominal value). Since the nominal displacement per magnet is ~140 mm, the good field region is required to be ±30 mm W x ± 12 mm H (1.18” W x 0.47” H) with respect to the magnet center lines.

The trajectory for the beam through the straight dipole magnet that is aligned in the tunnel with a ‘tilt’ with respect to the incoming beam direction of 3.28° and offset from the center line of ½ x a 35.2 mm (1.39”) sagitta is shown in Figure 2. The blue dots show the beam motion while the red dots show the magnet center line. The green dots represent the difference between the two and uses the right-hand scale on the plot. The purple and light blue lines which are parallel to the red center line represent the good field region.

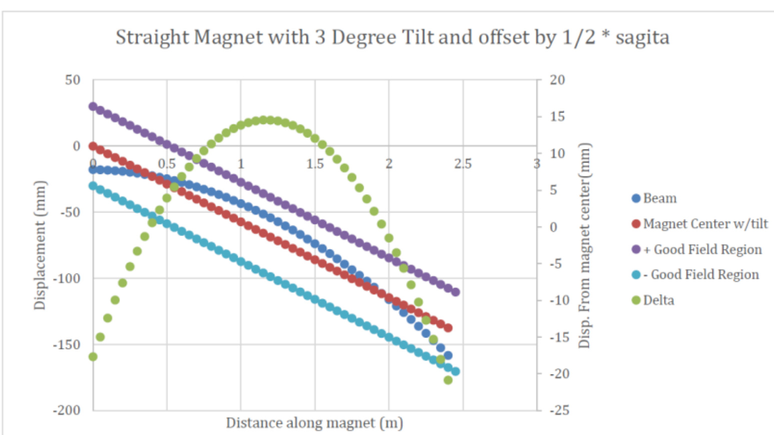

Figure 2: Beam Trajectories Through a Straight Regular Bend Dipole

The regular bend dipoles will be installed in the BTL between the Linac and Booster. In addition to the requirements in Section 2, the technical specifications for the regular bend dipoles are stated in Table 4.

Table 4: Regular Bend Dipole Technical Specifications

Requirement #	Requirement Statement
T-D-121.03.05-A001	The effective magnetic length shall be 2.34 m (92.13") at the maximum field
T-D-121.03.05-A002	Physical length including coils, water connections, and covers shall not exceed 2.6 m (102.4")
T-D-121.03.05-A003	The nominal gap shall be 52 mm ± 0.05 mm (2.047" ± 0.002") across the length of the magnet
T-D-121.03.05-A004	The good field region shall be ±30 mm wide x ± 12 mm high (±1.18" wide by ±0.47" high) with respect to the magnet center line
T-D-121.03.05-A005	The integrated field homogeneity in the good field region shall be ±0.1%
T-D-121.03.05-A006	The maximum field at the maximum current shall be 0.292 T with a nominal field at 800 MeV of 0.252 T
T-D-121.03.05-A007	The maximum current at the maximum field shall be ≤500 A
T-D-121.03.05-A008	The ramping rate shall be >10 A/sec
T-D-121.03.05-A009	The transverse dimensions including coils, water connections, and covers shall not exceed 0.8 m (31.5") in width and 0.41 m (16.14") in height.
T-D-121.03.05-A010	The maximum integrated field at the maximum current shall be 0.68 T-m with a nominal integrated field of 0.561 T-m and a nominal bend angle of 0.121 radians
T-D-121.03.05-A011	The coils shall be continuous (i.e., no splices)
T-D-121.03.05-A012	The magnets shall include provisions for alignment fiducials
T-D-121.03.05-A013	The magnet must accommodate a Fermilab designed 98.4 mm W x 50.0 mm H (3.87" W x 1.97" H) elliptical beam tube
T-D-121.03.05-A014	The maximum voltage per magnet at the maximum field shall be ≤ 15 V
T-D-121.03.05-A015	The magnet resistance at the nominal operating current shall be < 0.03 Ω
T-D-121.03.05-A016	All magnets are required to have an identical integrated BdL field within ±0.1% of each other
T-D-121.03.05-E003	The ends of the coils shall maintain a minimum clearance of 33 mm (1.3") from the ends of the beam tube transition plate

3.2. EOL Dipoles

There will be 2 vertical bend EOL dipoles installed in the Booster enclosure at the end of the BTL which are used for injection into the Booster. As with the regular dipoles in Section 3.1, the EOL dipoles are required to have an identical integrated BdL fields within ±0.1%. The EOL dipoles have a total vertical displacement of 210 mm. The technical specifications for the EOL dipoles are listed in Table 5. The maximum longitudinal length of the EOL dipole is 0.66 m. To accommodate this, FFDG will accept racetrack or saddle coil designs. FFDG recognizes the challenge of using a hollow, water-cooled conductor for meeting the physical dimensions specified in the table below. FFDG is open to considerations regarding various cooling solutions so long as the thermal specifications are met. The Booster enclosure is not temperature regulated so the design in which water cooling is not utilized must demonstrate the magnet does not exceed 46.1°C (115°F) when ambient conditions reach 37.8°C (100°F).

Table 5: EOL Dipole Technical Specifications

Requirement #	Requirement Statement
T-D-121.03.05-C001	The maximum effective magnetic length of 0.56 m (22.05”) at the maximum field
T-D-121.03.05-C002	Physical length including coils, water connections, and covers shall not exceed 0.66 m (26”)
T-D-121.03.05-C003	The nominal gap shall be 52 mm ± 0.05 mm (2.047” ± 0.002”) across the length of the magnet
T-D-121.03.05-C004	The good field region shall be ±14 mm x ±14 mm (±0.55” x ±0.55”) with respect to the magnet center line
T-D-121.03.05-C005	The integrated field homogeneity in the good field region shall be ±0.1%
T-D-121.03.05-C006	The maximum field at the maximum current shall be 0.387 T with a nominal field at 800 MeV of 0.203 T
T-D-121.03.05-C007	The maximum current at the maximum field shall be ≤ 110 A
T-D-121.03.05-C008	The ramping rate shall be >10 A/sec
T-D-121.03.05-C009	The transverse dimensions including coils, water connections, and covers shall not exceed 0.55 m (21.65”) in width and 0.4 m (15.75”) in height
T-D-121.03.05-C010	The maximum integrated field at the maximum current shall be 0.217 T-m with a nominal integrated field of 0.203 T-m and a nominal bend angle of 0.0416 radians
T-D-121.03.05-C011	The coils shall be continuous (i.e., no splices).
T-D-121.03.05-C012	The magnets shall include provisions for alignment fiducials
T-D-121.03.05-C013	The magnets shall accommodate for a 2” round beam tube
T-D-121.03.05-C014	The maximum voltage per magnet at the maximum field shall be ≤ 15 V
T-D-121.03.05-C015	The magnet resistance at the nominal operating current shall be < 0.09 Ω
T-D-121.03.05-C016	All magnets are required to have an identical integrated BdL field within ±0.1% of each other.

3.3. Regular Wide Dipoles

The regular wide dipole is the first dipole magnet in the BTL and requires special consideration. The magnet will contain a nominal 203.2 mm W x 44.45 mm H (8” W x 1.75” H) rectangular beam tube that accommodates: 1) a straight-ahead beam trajectory which directs the beam to the absorber when there is no current in the magnet and 2) a nominal bent trajectory which directs the beam to the rest of the BTL. The field uniformity for the deflected beam (i.e., nominal beam) which is center on the pole width still needs to meet the requirements. The integrated field strength in the straight-ahead beam tube shall be less than 0.01 T-m with no current flowing through the coils.

The regular wide dipoles are required to meet all of the technical specifications regular bend dipoles listed in Section 3.1 in addition to the technical specifications listed in Table 6. The design of this magnet could be the same as the regular dipoles.

Table 6: Regular Wide Dipole Technical Specifications

Requirement #	Requirement Statement
T-D-121.03.05-B001	The regular wide dipole shall have the same magnetic and electrical properties as the regular dipole
T-D-121.03.05-B002	The magnet design shall accommodate a 203.2 mm wide x 44.45 mm tall (8” x 1.75”) rectangular beam tube
T-D-121.03.05-B003	The integrated field for the straight-ahead beam trajectory shall be <0.01 T-m at 0 A current.

4. REVIEWS

To ensure a proper understanding of the specifications and the design proposals the subcontractor and FFDG shall conduct the following program reviews. The Subcontractor shall provide all documents required for each program review in electronic format.

1. Initial Kickoff Meeting
    a. The Initial Kickoff Meeting shall be held to review the contract requirements, deliverables, Subcontractor approach to executing the contract, and production schedule.

2. Preliminary Design Review (PDR)
    a. The PDR shall be held at the 60-80% design maturity state for each magnet type to ensure the design approach meets the technical requirements outlined in this SOW. Detailed designs are not expected, but preliminary design and analyses are required to demonstrate compliance with requirements. Presentations of the design, analyses, modeling and any early results are required. Supporting data and analyses for mechanical, power, thermal, and reliability assessments should be shown. Results from prototype testing, if available, should also be presented. The following deliverables are required for the PDR:
        i. Preliminary QC: Fabrication & Inspection Plan
        ii. Preliminary Design Report
        iii. Preliminary 3D models/drawings of all major components (i.e., core)
        iv. Preliminary Magnetic Measurement Plans
        v. Preliminary First Article Production Schedule

3. Final Design Review (FDR)
    a. The FDR shall be held at the 90-100% design maturity state for each magnet type to ensure the completed design achieves all physics, functional, technical, and interface requirements supported by engineering notes, drawings, etc. The technical areas addressed during the review include the overall final design as well as verification planning and compliance with the requirements. Approval of the FDR deliverables by FFDG is required before starting production of the first articles. The following Deliverables are required for the FDR:
        i. QC: Fabrication & Inspection Plan
        ii. Final Design Report~~s~~
        iii. Final 3D models and complete drawing sets of all magnets

iv. Final Magnetic Test Plan
v. Final Production Schedule

4. First Article Documentation Review and Testing
   a. The first articles shall undergo Factory Acceptance Tests (FATs) and SATs outlined in Section 8.1 and 8.2 respectively. As part of the deliverables for the FATs, the Subcontractor shall provide Final QC: Fabrication and Inspection Reports as well as Final Magnetic Measurement Test Reports for each first article. The documentation shall be reviewed and approved by FFDG prior to shipping. FFDG reserves the right to witness all first article FATs. Upon delivery, the first articles shall undergo SATs. Exercising the option for the production units is contingent on the first articles meeting the requirements in Section 3 as verified through the SATs outlined in Section 8.2.

5. Production Magnet Documentation Review and Testing
   a. The production units shall undergo FATs and SATs outlined in Section 8.1 and 8.2 respectively. As part of the deliverables for the FATs, the Subcontractor shall provide Final QC: Fabrication and Inspection Reports as well as Final Magnetic Measurement Test Reports for each production unit. The documentation shall be reviewed and approved by FFDG prior to shipping. FFDG reserves the right to witness all production unit FATs. Upon delivery, the production units shall undergo SATs as outlined in Section 8.2.

5. DELIVERY SCHEDULE AND HOLD POINTS

Table 7 outlines the delivery schedule for the deliverables and hold points required by the contract. The schedule represents the number of weeks After Receipt of Order (ARO). The Subcontractor is requested to submit a production schedule for review and approval by FFDG as part of the deliverables for the reviews as stated in Section 2. The production magnets shall be delivered in 3 batches. Other delivery schedules where fewer magnets are delivered per batch is allowed as long as quality is not sacrificed. As magnets are received, FFDG shall perform the SATs outlined in Section 8.2 to ensure the magnets the specifications in this SOW. Magnets that fail these tests shall be rejected and returned to the Subcontractor for repairs at the Subcontractors expense. To decrease time between deliveries of the production batches, the Subcontractor is permitted to proceed with fabrication for the next subsequent production unit batch prior to the completion of the SATs.

Table 7: Overall Project Schedule and Hold Points

Description	Weeks for Item	Weeks ARO
Kickoff Meeting	1	1
PDR Hold Point – FFDG review and approval of PDR deliverables	8	9
FDR Hold Point – FFDG review and approval of FDR Deliverables	8	17
First Articles (3 units) Documentation Review Hold Point – FFDG review and approval of QC: Fabrication & Inspection Report and Magnetic Test Report per magnet for approval to ship	12	29
Delivery of First Articles	6	35
First Articles Hold Point - SATs	4	39
First Article Acceptance – Approval of first articles and exercise option for production units	1	40
Production Units Batch 1 (15 units) Documentation Review Hold Point – FFDG review and approval of QC: Fabrication & Inspection Report and Magnetic Test Report per magnet for approval to ship	13	53
Delivery of Production Units Batch 1 (15 units)	6	59
Production Units SATs	4	63
Production Units Batch 2 (15 units) Documentation Review Hold Point – FFDG review and approval of QC: Fabrication & Inspection Report and Magnetic Test Report per magnet for approval to ship	13	76
Delivery of Production Units Batch 2 (15 units)	6	82
Production Units SATs	4	86
Production Units Batch 3 (14 units) Documentation Review Hold Point – FFDG review and approval of QC: Fabrication & Inspection Report and Magnetic Test Report per magnet for approval to ship	13	99
Delivery of Production Units Batch 3 (14 units)	6	105
Production Units SATs and Contract Closeout	4	109

6. REPORTING/MEETING REQUIREMENTS

The subcontractor shall submit a written subcontract status report to FFDG (i.e., in bullet form via email) monthly. Virtual Bi-weekly or monthly meetings shall also be held between the Subtractor and FFDG on an as needed basis to discuss project status.

The Subcontractor shall provide access to the subcontractor's manufacturing facility, which will include observation of the in-process work/testing as defined in this SOW. FFDG reserves the right to witness the SATs listed in Section 8.1. The Subcontractor is required to notify FFDG when the SATs will begin with a minimum of 35 days written notice.

7. QUALITY REQUIREMENTS

As part of QC, the Subcontractor is required to develop documentation as part of the deliverables in Section 2 to ensure the magnets meet the technical requirements listed in Section 3. The Subcontractor is required to procure, build, test, and deliver the magnets per the requirements listed in Section 3. The subcontractor shall immediately notify FFDG of any conflict or ambiguity between the specifications, SOW and/or drawings prior to any work being performed or cost being incurred by FFDG. In the event of a conflict, the order of precedence is as follows:

- Scope of Work
- Specifications
- General Terms and Conditions as identified in the Request for Proposal (RFP) documents
- Subcontractor's RFP response

The documentation listed below is considered a contract deliverable and is required for review and approval by FFDG prior to the start of fabrication, prior to shipment of the magnets, and/or acceptance of the magnets. In addition, the vendor is required to develop and maintain a production schedule. The production schedule shall outline the timeline for the execution and completion of tasks and deliverables described by this SOW. At a minimum, it shall include key project phases, milestone dates, production and fabrication activities, and delivery deadlines. The schedule ensures that all work is carried out in a timely and coordinated manner, allowing for regular review and adjustments as needed. Deviations from the schedule must be communicated and approved by FFDG prior to implementation.

- Material Certifications of Major Components: Material certification documents for the copper conductors and laminated steel used to fabricate the laminations shall be provided to FFDG to ensure they are in compliance with the specifications. The material certifications should include, but are not limited to, type of material, mechanical and chemical properties, lot or heat number, and the specifications the material complies with (i.e., ASME, ASTM, etc.).

- Final Design Report: The Subcontractor shall prepare and submit a comprehensive Final Design report documenting the complete design, analysis, and fabrication approach for each magnet type. The report must detail the magnetic design, including yoke and pole geometry, coil configuration, material selection, and field strength and quality analysis supported by 2D and 3D simulation results. The report must provide an overview of the mechanical design addressing analysis or calculations related to the structural integrity, thermal parameters, provisions for rigging, and tipping calculations. Furthermore, an overview of the electrical design is also required. The report shall also describe the general manufacturing and assembly process. The Final Design Report will serve as the basis for proceeding to procurement of materials and the start of fabrication.

- QC: Fabrication & Inspection Plan/Report: The supplier shall develop a QC plan for review and approval by FFDG prior to the start of fabrication. The QC plan shall include:
    - Description of equipment and testing/inspection methods planned to ensure requirements outlined in this SOW are met
    - Description of documentation planned to communicate contract conformance to FFDG
    - Description of nonconformance management process to be used

The QC plan shall outline in-process inspections for major manufacturing and fabrication steps including-core assembly, coil winding and potting, and final magnet assembly. The plan must define critical processes, specify dimensional tolerances and identify QC hold points. The plan shall describe all inspections and acceptance tests to be performed on the completed magnets. This shall include, but is not limited to, electrical tests (Hipot, resistance, etc.), mechanical dimensional tests, and hydrostatic measurements. The test plan shall specify test setups/equipment, measurement methods, data recording, and acceptance criteria. Any deviations from expected results must be clearly reported along with proposed corrective actions. The Subcontractor is responsible for maintaining detailed records throughout the fabrication and testing process and shall submit a final QC: Fabrication & Test Report that includes as-built data, non-conformance reports (if any), and complete test results. These reports shall contain the signature and title of the authorized personnel who perform the tests and measurements. The final Fabrication & Test Report shall be completed for all magnets and submitted to FFDG for review and approval prior to shipment.

- <u>Magnetic Measurement Test Plan/Report</u>: The Magnetic Test Plan that outlines the procedure, equipment, and criteria for verifying the magnetic performance of the magnets. The plan shall define the scope of testing, including the magnetic parameters to be measured, such as field strength, field uniformity, integrated field, and harmonics. The plan shall describe the test setup in detail including the type of instrumentation used (hall probes, rotating coils, etc.) and the post measurement processing of magnetic data results. The Subcontractor is responsible for maintaining detailed records throughout magnetic testing process and shall submit a final Magnetic Test Report that includes measured values for all key magnetic parameters. Any observed deviations or nonconformances shall be clearly identified, along with proposed or implemented corrective actions. Data shall be presented in both tabular and graphical form where appropriate and compared directly to the specified performance requirements. These reports shall contain the signature and title of the authorized personnel who perform the tests and measurements. The Final Magnetic Measurement Test Reports shall be completed for all magnets and submitted to FFDG for review and approval prior to shipment

8. ACCEPTANCE

8.1. Subcontractor Factory Acceptance Test

The Subcontractor shall complete all FATs listed in Table 8 at their facility and complete the required documentation as described in Section 7. The Subcontractor must have the capability to complete the General Inspection and Magnetic Measurement listed in Table 8 at their facility. The FATs shall be performed on 100% of magnets with the results attached to each magnet upon delivery. The test results must demonstrate that the magnets meet all technical specification requirements described in this SOW. The Subcontractor is permitted to conduct additional tests and/or suggest variations to the FATs in accordance with their standard practices and procedures for review and approval by FFDG.

Table 8: Subcontractor FATs

General Inspection		
Test Name	**Test Description**	**Acceptance Criteria**
Visual	Magnet assembly visual inspection for damage	Magnets must be free of damage, dirt, grease, oil, and chips
Dimensional	Measure the gap across the length of the magnet	52 mm ± 0.05 mm (2.047" ± 0.002")
Electrical	Hipot between coils and core at 1.5 kV	<5 µA Leakage Current
Electrical	Measure magnet total DC resistance	< 0.03 Ω
Electrical	Measure magnet inductance at 100 Hz and 1 kHz	Reference Only
Electrical	Ring Test to 100 V	No Shorts
Hydro Flow Rate	Measure water flow rate at the nominal pressure drop	<±5% of Calculated Nominal Flow Rate
Hydrostatic Leak	Pressure test water circuits at 15.5 Bar (225 psi) for 30 minutes	No Leaks

Magnetic Measurements	
Test Description	**Acceptance Criteria**
Magnetic Excitation	The nominal and maximum magnetic field shall be measured up to the maximum current as specified in Section 3
Integrated Field Gradient	All Integrated Field Gradients shall be within 0.1% of the nominal for each magnet type as specified in Section 3 at the maximum current
Field Homogeneity	The integrated field homogeneity in the good field region shall be ±0.1% for each magnet type as specified in Section 3
Transfer Function	The transfer function of the series of magnets operating at the nominal operating current should be <0.1% magnet to magnet.
Water Temperature	The outlet water temperature rise shall not exceed 10°C (18°F) above the nominal inlet temperature
Magnet Temperature	The temperature of all components shall be <46.1°C (115°F)

8.2. FFDG Site Acceptance Testing

FFDG will conduct the SAT's listed in Table 9 upon receipt of the magnets. All magnets need to meet the requirements as stated in this SOW as verified through acceptance testing. A magnet will not be accepted if any of these requirements are not met. All first articles and production magnets must pass incoming inspection which includes visual inspection for damage, dimensional verification, electrical inspection, and hydrostatic flow and leak measurements. In addition, magnetic measurements will be conducted on all first articles and a minimum of 10% of production magnets. Additional tests and measurements may be conducted but are not part of the SAT criteria.

Table 9: FFDG SATs

Incoming Inspection		
Test Name	**Test Description**	**Acceptance Criteria**
Visual	Magnet assembly visual inspection for damage	Magnets must be free of damage, dirt, grease, oil, and chips
Dimensional	Measure the gap across the length of the magnet	52 mm ± 0.05 mm (2.047" ± 0.002")
Electrical	Hipot between coils and core at 1.5 kV	<5 µA Leakage Current
Electrical	Measure magnet total DC resistance	< 0.03 Ω
Electrical	Measure magnet inductance at 100 Hz and 1 kHz	Reference Only
Electrical	Ring Test to 100 V	No Shorts
Hydro Flow Rate	Measure water flow rate at the nominal pressure drop	<±5% of Calculated Nominal Flow Rate
Hydrostatic Leak	Pressure test water circuits at 15.5 Bar (225 psi) for 30 minutes	No Leaks

Magnetic Measurements	
Test Description	**Acceptance Criteria**
Magnetic Excitation	The nominal and maximum magnetic field shall be measured up to the maximum current as specified in Section 3
Integrated Field Gradient	All Integrated Field Gradients shall be within 0.1% of the nominal for each magnet type as specified in Section 3 at the maximum current
Field Homogeneity	The integrated field homogeneity in the good field region shall be ±0.1% for each magnet type as specified in Section 3
Transfer Function	The transfer function of the series of magnets operating at the nominal operating current should be <0.1% magnet to magnet.
Water Temperature	The outlet water temperature rise shall not exceed 10°C (18°F) above the nominal inlet temperature
Magnet Temperature	The temperature of all components shall be <46.1°C (115°F)

8.3. Serialization

All units shall be assigned a unique serial number for identification and traceability. Each serial number shall be linked to the required documentation outlined in Section 7 as appropriate. The serial number shall be displayed on the nameplate as stated in Section 3. The serial number shall consist of a prefix designating the magnet type followed by a sequential number indicating the order in which the magnets were fabricated. The following shall be used as a template for the different magnet types:

- Regular Dipole – PIDA0XX
- EOL Dipole – PIDB00X
- Regular Wide Dipole – PIDC00X

9. SHIPPING

The magnets shall be cleaned and protected against corrosion prior to crating and shipment. This includes, but is not limited to, clearing the water circuits of residual water, protecting unpainted surfaces from corrosion (i.e., pole tips). Standard shipping methods and practices shall be used. The shipping crates shall be designed so that it can be moved using standard handling devices (i.e., overhead cranes, pallet jacks, etc.). All crates shall be clearly marked with contents with and weights including units. Irreversible indicators that record vertical shock loads of more than 10 G and tilt indicators shall be attached to the shipping crates. The packing shall be such that the magnets are not exposed to any environmental elements that will cause structural damage or cosmetic concerns. The packaging shall also prevent damage to any part of the magnet assembly.

10. Technical Requirements

The proposals shall be evaluated based on the following technical criteria in addition to the criteria in the RFP documents:

1. Fabrication Experience: Subcontractors are to include past construction experience for similar fabrication, and quality control completed on their premises. A list of 3 contracts completed in the past 5 years that have been awarded to the Subcontractor of approximately the same dollar magnitude as the FFDG proposal and are as similar as possible in nature to the requirements covered by this SOW. Specifically, the list should provide the following information for each contract:

    a) Contract Title
    b) Date of Award
    c) Contract Amount
    d) Client's Name and Address
    e) Client's Contact Information
    f) Description of Contract Deliverables with photos of devices (if shareable)

2. Infrastructure and Capabilities: Subcontractors are to outline their in-house infrastructure related to this specific job including, but not limited to, manufacturing and fabrication capabilities, magnetic testing, and general quality assurance program/equipment. The Subcontractor shall provide, at a minimum, the following information:

    a) Description of planned manufacturing and fabrication equipment to be used to meet the deliverables in Tables 1 through 6
    b) Magnetic testing infrastructure directly to the magnetic tests stated in Table 8
    c) Outline of the general Subcontractor QA program including certifications
    d) Outline of the Subcontractor QC program and planned equipment to be used directly related to the deliverables outlined in this SOW and described in Section 7

3. Preliminary Design: Subcontractors are to include their preliminary magnetic and mechanical designs for all the various types of dipole magnets outlined in this SOW. The preliminary mechanical designs shall, at a minimum, show and describe the overall design and fabrication procedures respectively. Detailed mechanical designs are not required. The preliminary magnetic

designs shall, at a minimum, show the designs that meet the specifications listed in Section 3. Specifically, the preliminary designs for all types of magnets must show the following:

a) Physical parameters outlined in Table 4, Table 5, and Table 6 have been met including the weight limitation in Table 3
b) Description of the core fabrication procedure and planned tooling to be used
c) Description of the coil winding process and the planned tooling to be used
d) Description of the coil epoxy impregnation process and planned tooling to be used as well as the type of epoxy
e) Simulations and graphical presentations showing magnetic and electrical requirements in Table 4, Table 5, and Table 6 have been met
f) Calculations describing the required cooling characteristics during full excitation/ operation according to the requirements in Table 3

4. Schedule: Subcontractors are to include a computer-generated Gantt chart (or similar) showing all activities and detailed schedules for all deliverables in Table 1 and Table 2. The schedule shall outline the timeline for the execution and completion of tasks and deliverables described by this SOW. At a minimum, it shall include key project phases, milestone dates, production and fabrication activities as well as delivery deadlines for the deliverables in Table 1 and Table 2.

11. POST-AWARD MANUFACTURER'S EXCEPTION DOCUMENT

If the Subcontractor requests a post-award change to the contract requirements (specifications, drawings, etc.), the Subcontractor shall provide to FFDG a Manufacturers Exception Document (MED). This document shall list all proposed changes. When drawings are affected, each drawing change shall be listed and include the drawing number, changed drawing values, and drawing location. FFDG must sign and return the MED prior to implementing changes. All changes shall be pursuant to and consistent with Clause 5 – CHANGES AND MODIFICATIONS as identified in the Terms and Conditions provided in the RFP documents.

12. Attachments

Drawing Number	Rev	Description
F10198049	A	BTL DIPOLE MAGNET BEAMTUBE WELDMENT
F10198047	A	BTL DIPOLE MAGNET ELLIPTICAL BEAMTUBE
F10218930	A	BTL WIDE APERTURE DIPOLE MAGNET BEAMTUBE WELDMENT
F10204063	B	BTL WIDE APERTURE DIPOLE MAGNET BEAMTUBE CHAMBER WELDMENT

APPENDIX A

A.1 ACRONYMS

ARO	After Receipt of Order
AR	As Required
BAL	Beam Absorber Line
BTL	Beam Transfer Line
EOL	End of Line
FATs	Factory Acceptance Tests
FDR	Final Design Review
FFDG	Fermi Forward Discovery Group, LLC
FNAL	Fermi National Accelerator Laboratory
LCW	Low Conductivity Water
Linac	Linear Accelerator
MED	Manufacturers Exception Document
MNPT	Male National Pipe Thread
PDR	Preliminary Design Review
PIP-II	Proton Improvement Plan Phase 2
QC	Quality Control
RFI	Request For Information
RFP	Request For Proposal
SATs	Site Acceptance Tests
SMR	Spherically Mounted Retroreflector
SOW	Statement of Work